

Structure-predicted solvent responses enable simulation-free hydration free-energy prediction

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Abstract

Hydration free energy reports how a molecule exchanges the gas phase for water, but accurate calculation can require explicit-solvent sampling and nuclear-quantum treatment for every new solute. We asked whether physical calculations could instead define reusable response coordinates that are learned from molecular structure. SolvAI first predicts 15 solvent-response priors learned from benchmark-disjoint quantum-continuum, alchemical, empirical and conformational data, then combines these priors with molecular fingerprints and descriptors in an experimentally supervised hydration model. A new prediction begins from a SMILES string and runs no molecular dynamics, path-integral dynamics or probe calculation. In a preregistered matched five-fold comparison on the 85 neutral solutes used to establish ARROW, response priors reduce mean absolute error from 0.303 to 0.202 kcal mol⁻¹ (paired change -0.101; 95% bootstrap interval -0.215 to -0.020), close to the reconstructed ARROW/PIMD8 error of 0.205 kcal mol⁻¹. Five complete split repeats give 0.207 ± 0.004 kcal mol⁻¹. Shuffling the priors abolishes the gain, whereas the advantage survives global scaffold, family, cluster and similarity exclusions. These results show that aligned physical responses can be amortized into structure-only molecular prediction, while identifying chemical extrapolation and high-fidelity response inference as the remaining limits.

Introduction

Hydration free energy is the reversible work of transferring a molecule from the gas phase into water. It connects molecular structure to partitioning, binding and reactivity, and is a stringent test of how a model balances electrostatics, polarization, dispersion, repulsion and solvent reorganization. Experimental values are available for many small neutral compounds, notably through FreeSolv, but predicting them with comparable precision remains difficult.^{1,2}

Physical simulation approaches the problem by resolving the solvent response. In an alchemical calculation the solute–water interaction is varied along a coupling path, and the resulting ensemble averages are integrated or combined statistically.³ More detailed Hamiltonians improve the intermolecular interaction; path-integral molecular dynamics (PIMD) can additionally represent nuclear delocalization.⁴ These ingredients can be accurate, but they demand repeated sampling for each new molecule. The ARROW study made this accuracy–cost relation unusually sharp: on 85 neutral solutes, classical ARROW has an exactly reconstructed mean absolute error (MAE) of 0.785 kcal mol⁻¹, whereas eight-bead PIMD reduces it to 0.205 kcal mol⁻¹.⁵ Subsequent learned-potential work reproduced ARROW water-pair interactions and quantum corrections within simulation, showing that parts of this description are learnable while retaining trajectory-based inference.^{6,7}

Molecular machine learning usually takes a different route. A graph, fingerprint or three-dimensional structure is mapped directly to a thermodynamic endpoint. Models can be pretrained on quantum calculations or corrected across levels of theory, and transfer from computed to experimental solvation has proved useful.^{8–11} Yet a direct endpoint model is not explicitly asked to reconstruct the physical responses that determine transfer into water. Conversely, a simulation surrogate often reproduces one expensive calculator and inherits its target and domain.

Here we test a third arrangement. Physical calculations are used to define compact, reusable response coordinates. Structure-to-response models learn those coordinates across external data; a second model uses their predictions, together with molecular structure and experimental hydration labels, to learn the endpoint (Fig. 1). The resulting system, SolvAI, still contains learned response models at inference, but their expensive source calculations are not rerun. PIMD-derived features were tested during development and were not retained. PIMD8 therefore serves as a demanding accuracy reference, not as a teacher of the final model.

The central test is not whether an elaborate model outperforms a weak baseline. We compare SolvAI with the same endpoint architecture, experimental labels, molecular representation, sample weights, folds and random seeds after removing only the 15 predicted response priors. We further destroy molecule–response alignment by shuffling those priors, exclude related chemistry from the complete endpoint training pool, and remove all ARROW experimental labels. Together these controls ask whether predicted physical response contributes transferable information beyond ordinary structure regression.

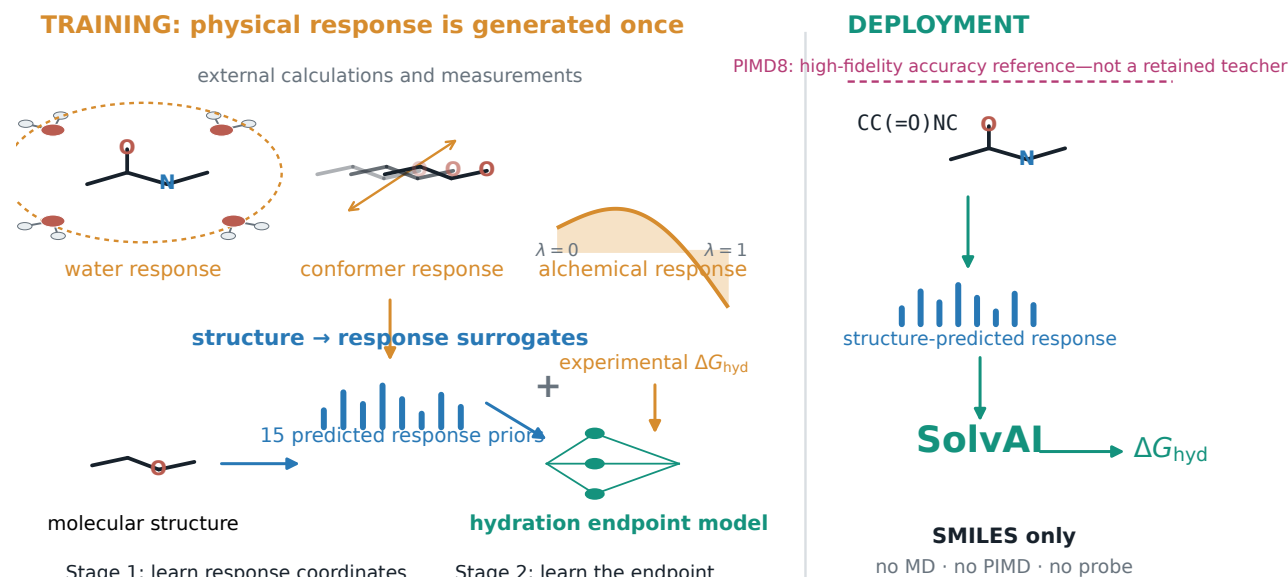


Figure 1: Physical response becomes reusable molecular supervision. External calculations and measurements describe complementary aspects of transfer into water, including continuum response, conformational reweighting and alchemical interaction response. In stage 1, structure-to-response surrogates learn 15 compact coordinates. In stage 2, their predictions and molecular descriptors are fitted to experimental hydration free energies. Deployment repeats only the learned mappings: a SMILES string produces a hydration free energy without MD, PIMD or a probe calculation. ARROW/PIMD8 is shown separately as an accuracy reference and is not a retained teacher.

Results

Reusable solvent-response priors

The final response vector spans distinct approximations to solvation. One coordinate describes COSMOtherm water response from CombiSolv-QM; five Abraham axes encode polarizability, hydrogen-bonding and partition response; two coordinates are residual-corrected explicit- and implicit-solvent predictions; one represents SMD(water); and six summarize conformer reweighting and the distribution of water response in ConfSolv.^{9;12–17} Every coordinate is generated by a surrogate from structure. The endpoint receives a 2,048-bit Morgan fingerprint, 217 RDKit descriptors and these 15 predictions.

The response sources differ deliberately in scale and physical resolution. MolSolv contributed 1,729,545 SMD(water) calculations, of which 350,359 structures remained in the confirmatory teacher after quality and identity filtering.¹⁸ ConfSolv contributed 5,392,567 water conformer records; aggregation, target availability and standardized-identity filtering left 17,829 structures for the response model. These counts are not placed on a common axis because one source counts calculations and the other conformer records. Complete definitions appear in Supplementary Table 1 and source-specific provenance in Extended Data Fig. 2.

An expanded identity audit changed the original release analysis. Exact connectivity had already been removed from every supervised external source, but conservative fragment, uncharging and canonical-tautomer standardization found 2 equivalent CombiSolv-QM records, 32 MolSolv records and 22 ConfSolv records. We removed them, refitted the three affected teachers while preserving all remaining source split memberships, and repeated every endpoint control reported below. The 1,280-label experimental endpoint pool contained no exact or standardized benchmark identity. This corrected analysis is the sole source of the main numerical results.

A matched test of physical information

Removing only the predicted responses from the frozen endpoint gives an OOF MAE of 0.303 kcal mol⁻¹ (Fig. 2a). Adding the compact response block lowers the error to 0.212; adding SMD(water) gives 0.207; and the complete stack reaches 0.202 kcal mol⁻¹. The last value lies on the same accuracy scale as ARROW/PIMD8 (0.205 kcal mol⁻¹), although the paired comparison does not establish superiority over PIMD8.

The response advantage is resolved at molecule level. Full SolvAI improves the absolute error for 53 of 85 solutes and reduces MAE by -0.101 kcal mol⁻¹ relative to the matched endpoint (95% paired-bootstrap interval, -0.215 to -0.020; Fig. 2b). The empirical and residual-corrected block is independently positive. Computation-derived, SMD and ConfSolv blocks are weaker in isolation, but their complete combination is positive (Fig. 2c). This is evidence of complementary response information, not an additive decomposition of hydration free energy.

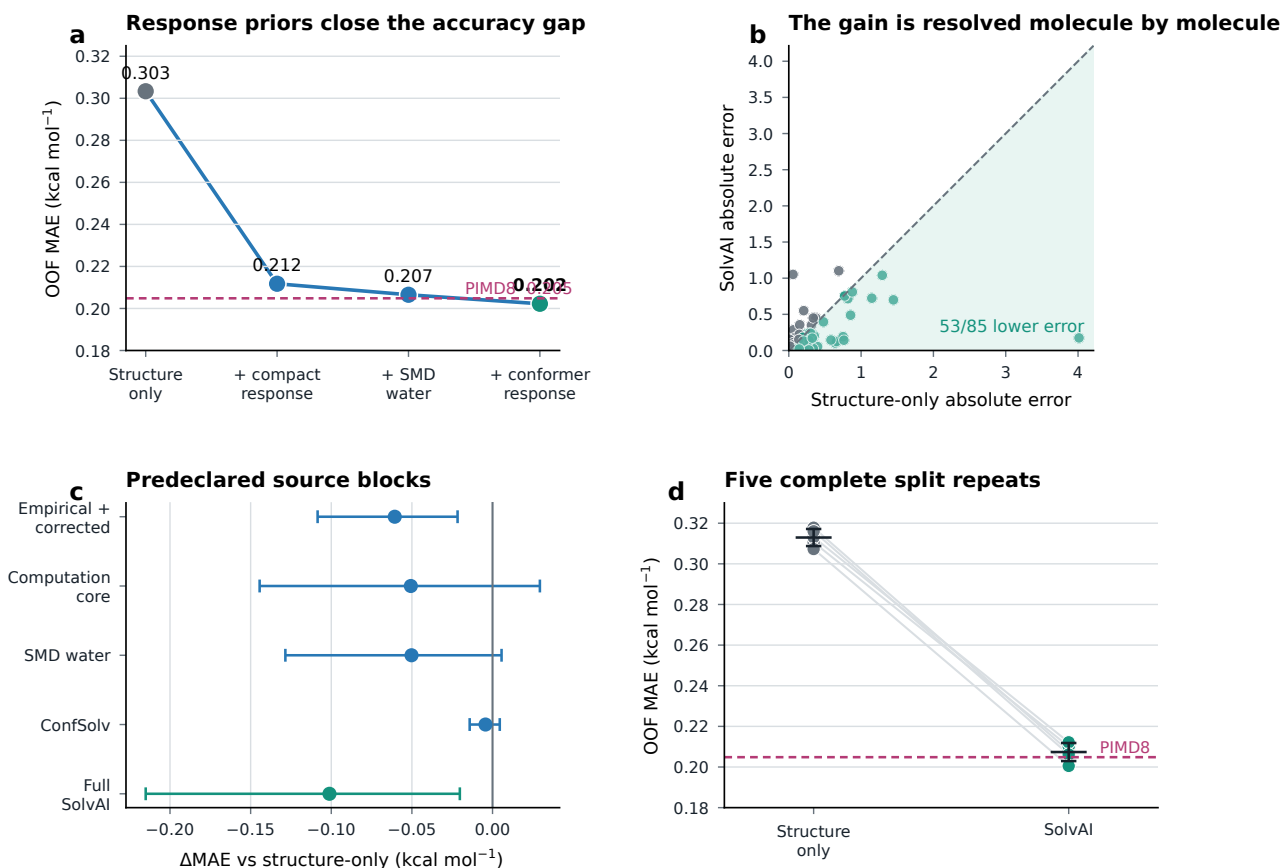


Figure 2: Molecule-aligned response priors improve a matched hydration endpoint. **a**, Predeclared fixed-partition progression. Each endpoint uses the same experimental labels, descriptors, architecture, weights, folds and seeds; only the response block changes. The dashed magenta line marks ARROW/PIMD8 on the same 85 solutes. **b**, Paired absolute errors for the matched structure-only endpoint and full SolvAI. Shading marks lower SolvAI error. **c**, MAE change for predeclared response blocks relative to structure only; bars are 95% molecule-bootstrap intervals. **d**, Paired results for five complete split repeats. Thin lines join methods within the same partition; black marks show mean $\pm$ sample s.d.

Arbitrary additional columns do not reproduce the result. Across five independent permutations of the 15-column block, the fixed-partition mean is 0.306 kcal mol⁻¹, close to the 0.303 matched control and far from aligned SolvAI. The paired difference between aligned and mean-shuffled priors is -0.104 kcal mol⁻¹ (95% interval, -0.214 to -0.025). Alignment between a molecule and its predicted response is therefore necessary for the observed gain.

Partition choice changes the absolute point estimate but not the controlled advantage. Across five complete five-fold partitions, matched structure-only prediction gives 0.313 kcal mol⁻¹ on average, whereas SolvAI gives 0.207 ± 0.004 kcal mol⁻¹; SolvAI is better in all five partitions (Fig. 2d). The appropriate headline is PIMD8-level accuracy on this reference chemistry, not uniformly sub-0.20 performance.

Transfer beyond nearby labels

Random molecular folds can leave close chemical neighbours in the external endpoint pool. We therefore applied each separation rule to all endpoint-supervised molecules, not only the 85 reference solutes. For each outer test partition, public and ARROW training rows in the held-out family, scaffold or molecular cluster were removed. A complementary analysis excluded every training row at or above a prescribed Morgan similarity to any test molecule.

The response-prior advantage survives all predeclared regimes (Fig. 3). At a maximum allowed train-test similarity of 0.70, MAE falls from 0.343 to 0.211 kcal mol⁻¹; cluster separation lowers it from 0.349 to 0.216. Scaffold and functional-family separation are much harder, but the same comparison favours SolvAI: 0.728 to 0.376 and 1.262 to 0.468, respectively. Every paired 95% bootstrap interval lies below zero. Response priors therefore retain value away from nearby endpoint labels, even where absolute endpoint accuracy is not yet sufficient.

We next removed all 85 ARROW experimental labels from endpoint training. Using only the same 1,280 external labels, the matched structure model gives 0.385 kcal mol⁻¹ and SolvAI gives 0.257 kcal mol⁻¹

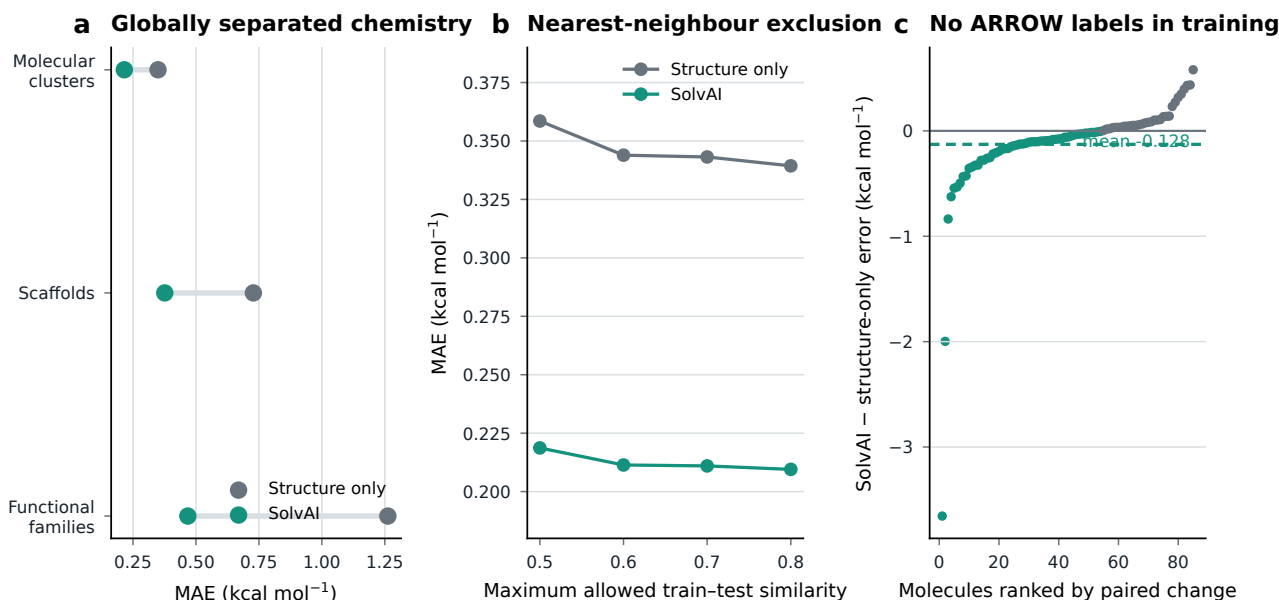


Figure 3: The response-prior advantage survives chemical separation. **a**, Matched MAEs after removing every endpoint training label in the held-out molecular cluster, scaffold or functional family. **b**, Sensitivity exclusion removes every training row at or above the indicated maximum Morgan similarity to a test molecule. **c**, Molecule-level change in absolute error when no ARROW experimental label is used for training; negative values favour SolvAI. All response teachers use the standardized-equivalent exclusions described in Methods.

(paired change -0.128 ; 95% interval, -0.246 to -0.037). This transport test is not an independent external benchmark—the method was developed in the context of ARROW chemistry—but it shows that response priors retain value without fold-local adaptation to the reference set.

What makes physics transferable

The benefit is not a generic consequence of adding physically motivated outputs. Compact ConfSolv response summaries transfer better than graph or feed-forward latent representations trained on the same source (Fig. 4a). Other candidate signals—including additional force-field diagnostics, pair-interaction data and learned implicit-solvent hierarchies—were non-improving in matched campaign screens (Extended Data Fig. 3). These results do not rank the underlying physical methods. They test whether each representation can be inferred from structure accurately enough, and with sufficiently non-redundant information, to improve the endpoint.

Short PIMD2 calculations at coupling values $\lambda = 0.1, 0.5$ and 0.9 make this constraint explicit. Structure-only heads captured broad response trends, but component MAEs ranged from 1.27 to 5.20 kcal mol⁻¹ (Fig. 4b). Adding the predicted PIMD2 response worsened the matched campaign endpoint from 0.196 to 0.201 kcal mol⁻¹; a predicted classical-NQE-PIMD hierarchy reached 0.215 , and their combination 0.219 (Fig. 4c). Direct integration of the predicted response curve yielded 1.51 kcal mol⁻¹ (Extended Data Fig. 5). No PIMD-derived coordinate is present in the final model.

The experiments suggest a three-part transfer condition. A response must be relevant to the endpoint, learnable from the deployed molecular representation, and non-redundant with the other priors. Broad physical fidelity alone does not guarantee useful distillation. Here, compact solvent-response coordinates learned from much larger external sources satisfy that condition more effectively than sparse, high-fidelity alchemical targets.

Discussion

SolvAI changes which part of a physical calculation is amortized. It does not emulate one trajectory engine or inherit a single simulation value. Instead, several solvation calculations define a low-dimensional response vocabulary; structure surrogates reproduce that vocabulary; and an experimentally supervised endpoint learns how the predicted responses combine for hydration. The matched control and shuffled-prior experiment show that the benefit is not attributable to a larger feature matrix alone. Global separation and zero-ARROW transfer further show that it is not confined to close endpoint-labelled neighbours.

This arrangement is related to knowledge distillation, Δ -learning, multi-fidelity modelling and transfer learning from computed to experimental properties, but assigns the computed data a different role.^{8;9;19} Rather than learning one correction from a low-fidelity scalar to a high-fidelity scalar, SolvAI predicts a set of heterogeneous

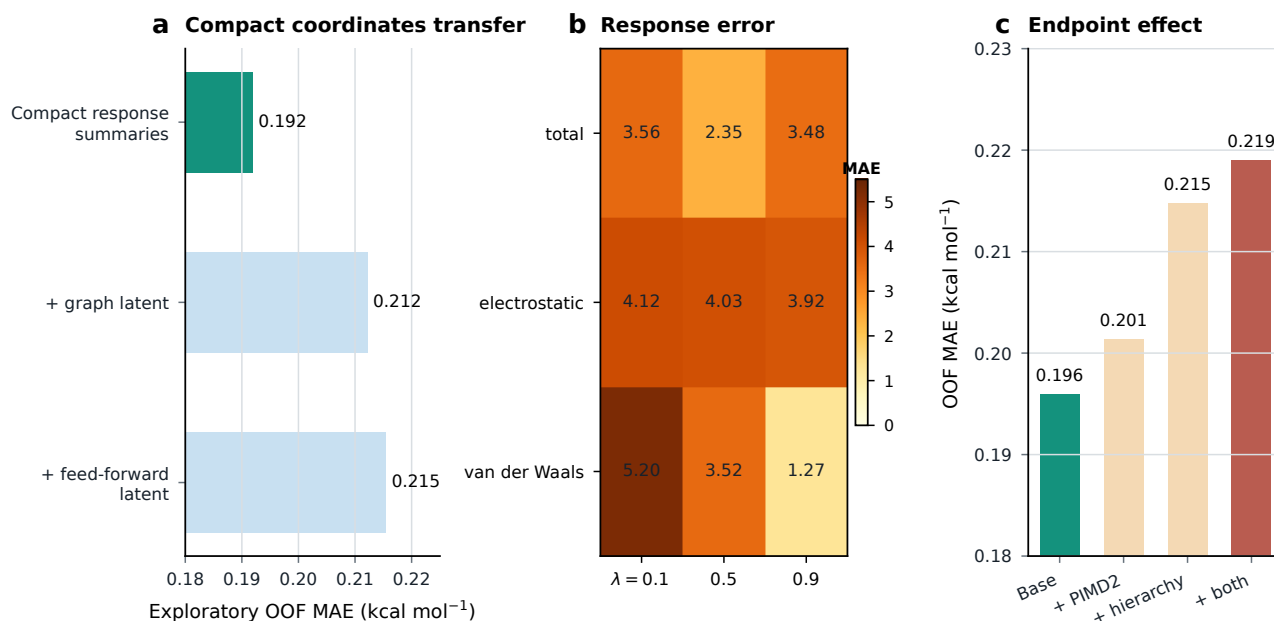


Figure 4: Transfer is limited by structure-to-response accuracy. **a**, Exploratory matched screen comparing compact ConfSolv summaries with graph and feed-forward latent coordinates from the same source. **b**, OOF error of structure-predicted PIMD2 total, electrostatic and van der Waals $dH/d\lambda$ responses. Values are kcal mol⁻¹. **c**, Consequence of adding those predicted high-fidelity responses to the matched campaign endpoint. These analyses explain a bottleneck; none of the PIMD-derived blocks is retained in SolvAI.

response coordinates and exposes them to a separate endpoint model. The response vector can therefore combine information from methods that differ in scope, fidelity and coverage. The successful object is not “simulation” in the abstract; it is an aligned, learnable representation of solvent response.

The evidence also sets a clear boundary. The primary reference contains only 85 neutral solutes, and the method was developed in its context. The confirmatory five-partition mean is 0.207 kcal mol⁻¹, so the evidence supports an accuracy scale rather than a threshold claim. Absolute error rises to 0.376 and 0.468 kcal mol⁻¹ when all endpoint labels sharing the test scaffolds or functional families are removed. The frozen model is not validated for ions, salts, biomacromolecules or substantially larger chemical space. Nor is this a generic FreeSolv ranking: 80 of the 85 reference identities occur in FreeSolv, and FreeSolv-related records contribute to endpoint training only after benchmark identity exclusion.

The negative high-fidelity result is informative. To distil alchemical response directly, the model needs enough protocol-matched examples to predict component curves at substantially better than the present 1.27–5.20 kcal mol⁻¹. A focused set of roughly 250–500 molecules spanning amides, aromatics, ethers, acids and alkanes, with complete λ -resolved total, electrostatic, polarization and dispersion/repulsion response and paired classical/PIMD values, would test that next step. It is a more specific requirement than simply collecting more endpoint labels.

The broader opportunity is to treat costly computation as scientific supervision with a lifetime beyond one query. Once a useful response is represented across enough chemistry, its information can be predicted from structure and reused at negligible marginal simulation cost. Hydration provides a demanding demonstration because the accuracy reference includes explicit solvent and nuclear quantum effects. The same principle may apply wherever physical computation exposes transferable response coordinates that a deployed model can learn.

Methods

Reference set and targets

The primary evaluation uses the 85 neutral organic solutes reported in the ARROW hydration analysis at 298 K.⁵ One water condition was retained per connectivity. The released benchmark table contains canonical structure, experimental hydration free energy, experimental uncertainty where available, classical ARROW, PIMD4 and PIMD8 values, and all fold assignments. The exact reconstructed PIMD8 MAE is 0.205 kcal mol⁻¹; the source article reports approximately 0.20 kcal mol⁻¹.

160 Endpoint labels and features

161 The external endpoint pool contains 1,280 neutral, benchmark-disjoint experimental hydration labels: 1,147
162 connectivity-resolved records from a FreeSolv/CombiSolv-Exp merge and 133 SoluteML dGsolvDB3 records
163 supported by at least two source measurements.^{2;13;20} Each endpoint input has a 2,048-bit radius-2 Morgan
164 fingerprint and 217 RDKit two-dimensional descriptors. Non-finite values are handled by a median imputer
165 fitted inside each endpoint fit.

166 Response surrogates

167 Fifteen response coordinates are generated by six artifact families (Supplementary Table 1). CHEMELEON-
168 initialized D-MPNNs predict CombiSolv-QM/COSMOtherm and MolSolv SMD(water) targets.²¹ ExtraTrees
169 models predict five Abraham axes and the physical and residual components of corrected OpenFF and GBn2
170 responses. LightGBM models predict six ConfSolv conformer and response-distribution summaries. The latter
171 are derived from the H₂O subset of ConfSolv. The final coordinate list, target units, source sizes, transformations
172 and inference calculations are given in Supplementary Table 1. PIMD-derived targets are absent from this list.

173 The endpoint is an ensemble of three **ExtraTreesRegressor** pipelines,²² each with 360 trees, maximum
174 feature fraction 0.7, minimum leaf size 2, no bootstrap and random seeds 11, 29 and 47. Predictions are averaged.
175 Public endpoint rows have sample weight 1 and available ARROW outer-training rows weight 3. The matched
176 structure-only model is identical except that the 15 response columns are removed.

177 Identity standardization

178 RDKit 2026.03.5 was used to generate canonical isomeric SMILES, full InChIKeys and connectivity blocks.²³
179 The confirmatory audit additionally applied MolStandardize cleanup, largest-fragment selection, uncharging
180 and canonical tautomer enumeration. Exact and standardized benchmark equivalents were removed from every
181 supervised external source. The audit found no such match in the endpoint pool. It identified 2 CombiSolv-QM,
182 32 MolSolv and 22 ConfSolv source records beyond the original exact-connectivity exclusion. The affected
183 teachers were refitted with all remaining molecules held in their original train, validation or test split. Full pair
184 lists, identifiers and split-verification hashes are released.

185 Confirmatory evaluation

186 The analysis protocol was committed before valid confirmatory model fitting. The primary partition is the
187 existing five-fold assignment on the 85 molecules. Five complete shuffled K-fold partitions use seeds 314159,
188 271828, 161803, 141421 and 173205. In each outer fold, all 1,280 external labels and the available ARROW
189 outer-training labels are fitted; the test target is absent. Feature blocks, ExtraTrees hyperparameters, sample
190 weights and model seeds are identical across all comparisons.

191 For the negative control, the 15 response columns are permuted as one block, thereby preserving their
192 joint distribution while destroying molecule alignment. Public, outer-training and outer-test response rows are
193 permuted separately within each fold. Five permutation seeds (88001–88005) are evaluated for the fixed partition
194 and each of the five repeats.

195 Chemical separation

196 Separation is applied to the complete endpoint-supervised pool. Functional families are assigned by a deterministic
197 SMARTS precedence; five outer folds are formed by group. Bemis–Murcko scaffolds are canonicalized, with
198 acyclic structures grouped by primary family. Taylor–Butina clusters use radius-2, 2,048-bit Morgan fingerprints
199 at Tanimoto similarity 0.70. For each outer test fold, every public or ARROW training row sharing a held-out
200 group is removed. A separate nearest-neighbour analysis removes every training row with similarity at or above
201 0.50, 0.60, 0.70 or 0.80 to any test molecule. No threshold was selected by performance. Exact SMARTS,
202 assignments, training counts and maximum retained similarities are released as Supplementary Data 3.

203 Zero-ARROW transfer

204 The frozen endpoint configuration was fitted once to the 1,280 external experimental labels without any ARROW
205 experimental value. Both matched structure-only and full response-prior models then predicted all 85 ARROW
206 molecules. This analysis measures transport without reference-set endpoint adaptation; it is not described as an
207 independent external test because model development used the reference chemistry.

Statistics

MAE in kcal mol⁻¹ is primary. RMSE, median absolute error and the fraction of molecules improved are secondary. All comparisons are paired by molecule. Ninety-five per cent percentile intervals use 100,000 molecule bootstrap resamples with seed 20260828. For repeated partitions, each molecule’s absolute error is first averaged across repeats before resampling. A predeclared comparison is positive when the complete paired interval for candidate-minus-control MAE is below zero, negative when above zero and neutral otherwise; an absolute difference of at least 0.010 kcal mol⁻¹ is additionally termed material. No multiplicity correction or post-result feature selection is used.

Deployment artifact

After evaluation was frozen, the response teachers were fitted with standardized benchmark equivalents removed and the endpoint was refitted on the 1,280 public plus 85 ARROW labels. This all-data artifact is used only for deployment, never for paper accuracy. Inference canonicalizes a SMILES string, computes descriptors, evaluates the six frozen response-surrogate artifacts and averages three endpoint regressors. It does not read an experimental value, family label, scaffold, fold identifier, trajectory or PIMD result. Software versions, hashes and a SMILES-only smoke test are reported in Supplementary Methods and Tables 6–7.

Data availability

All benchmark records, held-out predictions, split assignments, audit tables and derived source manifests required to reproduce the paper are provided in the SolvAI repository. Large third-party datasets are not redistributed where their size or licence makes that inappropriate; immutable URLs or DOIs, hashes, licences and exact preprocessing instructions are supplied in `repro/DATA_PROVENANCE.md`.

Code availability

Source code, the packaged inference artifacts, confirmatory analysis scripts, figure generation, manuscript source and a locked environment are available at <https://github.com/Scientific-Computing-Lab/SolvAI>. A frozen-artifact path reproduces predictions, metrics, figures and PDFs without rerunning physical calculations.

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Extended Data figure legends

Extended Data Fig. 1 | Molecule-level prediction and residuals. **a**, Confirmatory SolvAI OOF prediction against experiment. **b**, Signed residual against experimental hydration free energy. **c**, Distribution of the paired absolute-error change relative to the matched structure-only endpoint.

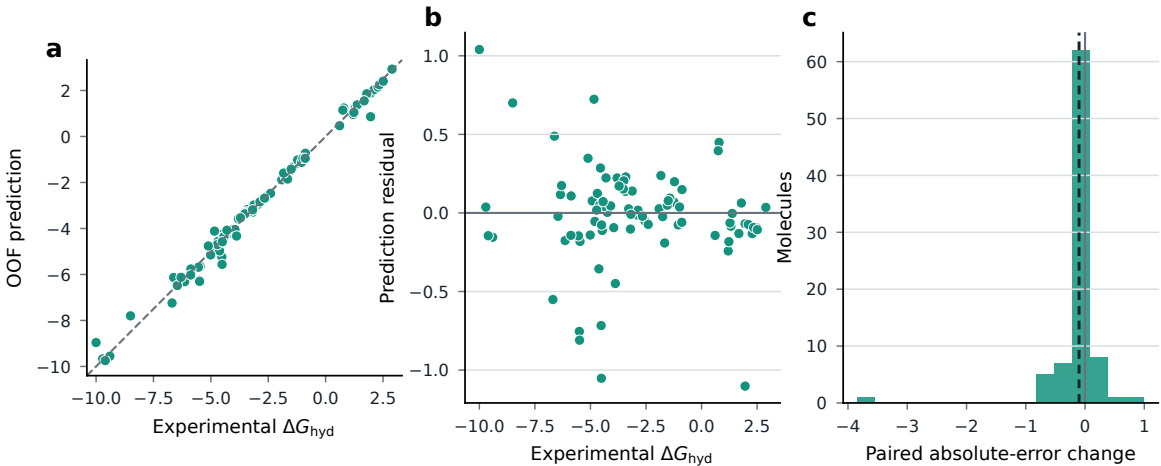
Extended Data Fig. 2 | Source-specific provenance and standardized identity exclusion. Counts are shown in their native source units and normalized only within source. The audit removed standardized benchmark equivalents before the confirmatory teacher refits; no endpoint experimental label was removed.

Extended Data Fig. 3 | Alternative physical-supervision routes. Exploratory fixed screens of additional diagnostic, latent and hierarchy features. The dashed line is the matched campaign response base for those screens. These experiments were not used to select the confirmatory controls.

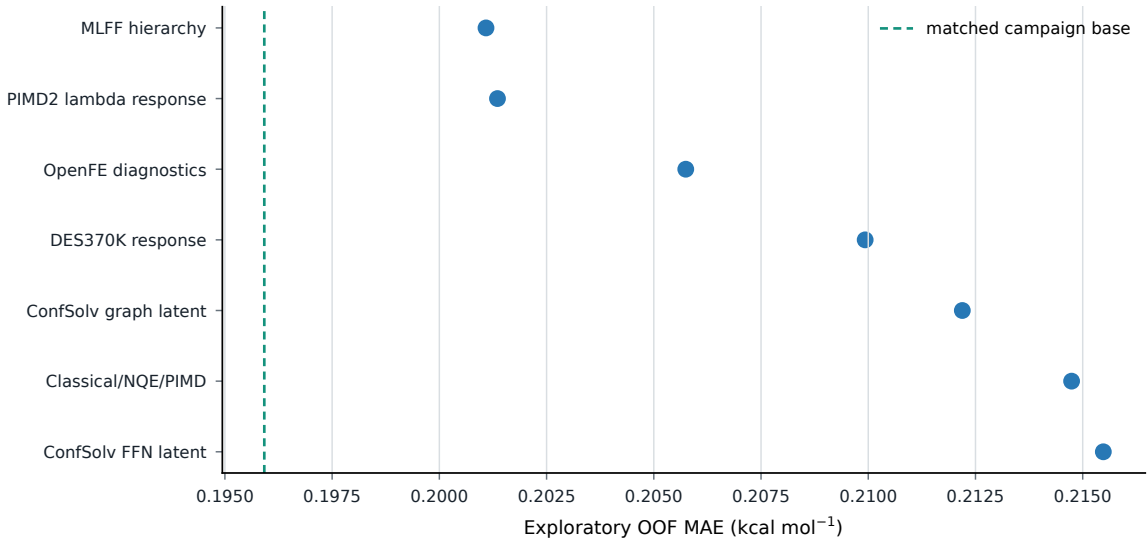
Extended Data Fig. 4 | Simulation-assisted reference. Accuracy–compute curves for selective PIMD policies from the discontinued simulation-assisted branch. They are reported for context and are not part of SolvAI, whose deployed path never routes to simulation.

Extended Data Fig. 5 | Multi- λ response experiment. **a**, OOF response-head error at three coupling states. **b**, Matched endpoint consequences and the separately scaled error obtained by direct integration of the predicted curve.

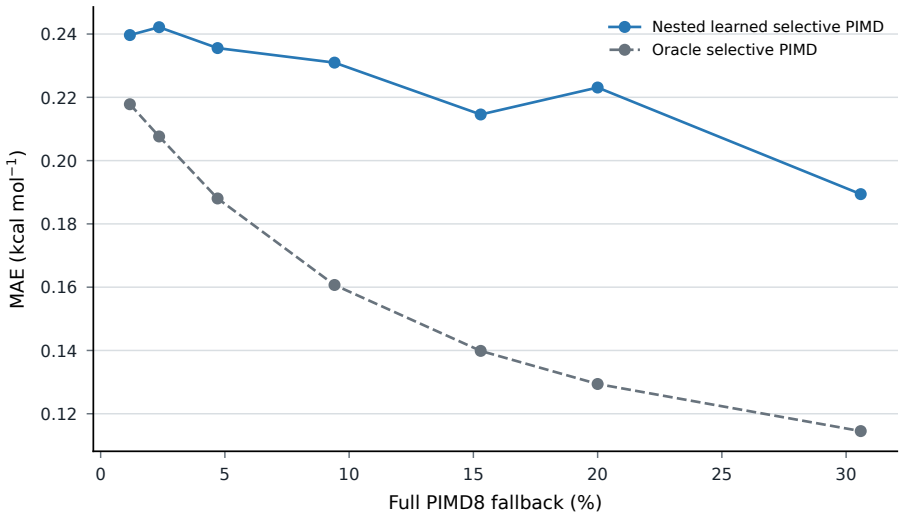
Extended Data Fig. 6 | Molecular and chemical-separation errors. **a**, Individual fixed-partition SolvAI errors grouped by diagnostic chemical family; ticks show family means and labels give sample sizes. **b**, Matched structure-only and SolvAI errors under four global separation regimes.

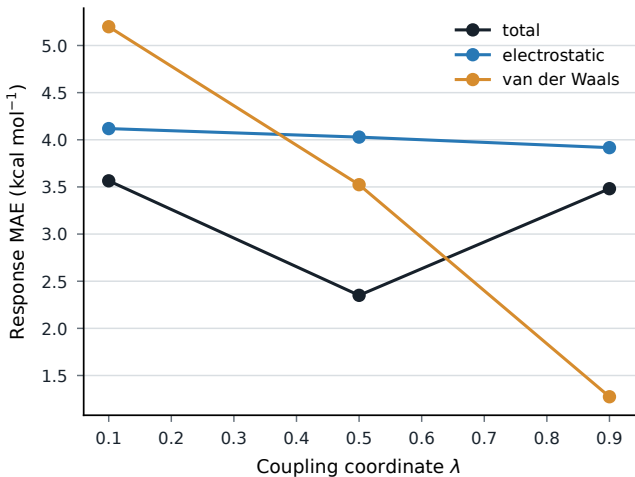
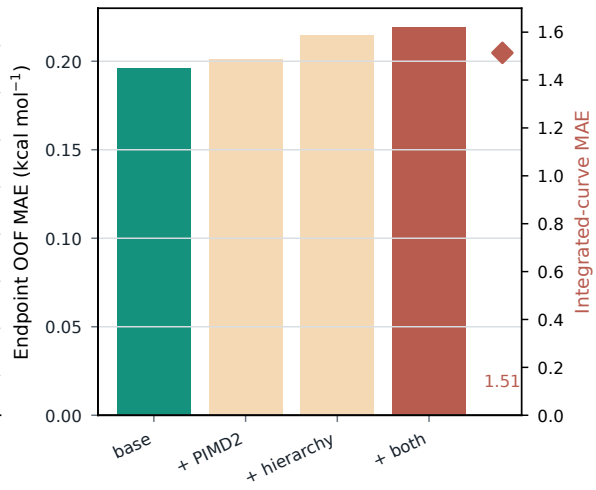


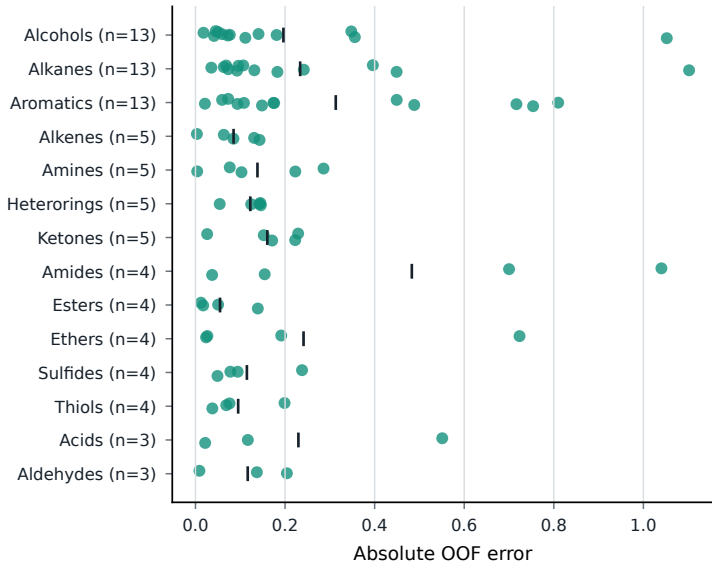
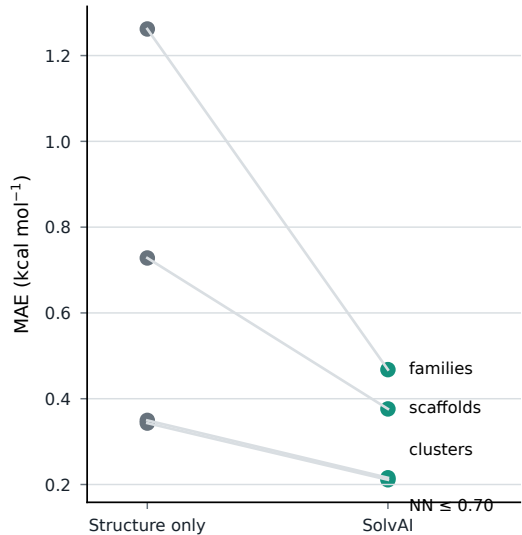




Simulation-assisted reference (not SolvAI)



a**b**

a**b**

Extended Data Table 1 — Final model and evaluation comparison

Method	Simulation at inference	Fixed OOF MAE	Global family MAE	Global scaffold MAE
Classical ARROW	yes	0.785	—	—
ARROW/PIMD8	yes	0.205	—	—
Matched structure-only	no	0.303	—	—
SolvAI	no	0.202	0.468	0.376
SolvAI, zero ARROW labels	no	0.257	—	—